

SMALLEST COMPONENTS

in PCBA Assembly

01005 • 008004 • 0201 • Micro-BGA • CSP • WLCSP

Size Comparison • Assembly Challenges • Process Requirements

PCBA Manufacturing & Miniaturization Standards

IPC/J-STD-001 | IPC-A-610 | IPC-7093 | IPC-7527

1. OVERVIEW OF MINIATURE COMPONENTS

Modern electronics demand ever-smaller components to enable portable, lightweight, and high-functionality devices. The PCBA industry has progressed from 1206 passives (3.2×1.6 mm) in the 1980s to **008004** (0.25×0.125 mm) in 2026 — a 650× reduction in footprint area. This guide covers the smallest components used in in-house PCBA assembly and the specialized processes required to handle them.

Evolution of SMD Component Sizes:

- **1980s:** 1206 (3.2×1.6 mm) — standard for early SMT
- **1990s:** 0805 (2.0×1.25 mm) — consumer electronics
- **2000s:** 0603 (1.6×0.8 mm) — compact devices
- **2010s:** 0402 (1.0×0.5 mm) — smartphones
- **2015s:** 0201 (0.6×0.3 mm) — high-density assemblies
- **2020s:** 01005 (0.4×0.2 mm) — wearables, medical implants
- **2026:** 008004 (0.25×0.125 mm) — next-gen microelectronics

Why Miniaturization Matters:

- **Space savings:** More functionality per cm^2 of PCB area
- **Weight reduction:** Critical for aerospace and portable devices
- **Improved electrical performance:** Shorter traces = lower parasitic inductance/capacitance
- **Cost efficiency:** Smaller PCBs = lower material costs at volume

■ **In-House Reality:** Most facilities (including in-house) currently handle 0201 and 0402 as standard. 01005 requires specialized equipment and processes. 008004 is emerging technology requiring jet dispensing, advanced vision systems, and specialized reflow profiles.

2. PASSIVE COMPONENT SIZE COMPARISON

2.1 Imperial vs. Metric Nomenclature

SMD component sizes use a 4-digit code representing dimensions in **hundredths of an inch** (Imperial) or **tenths of a millimeter** (Metric). This dual system causes frequent confusion.

Imperial Code (L × W in 0.01 inch):

- **01005**: 0.01×0.005 inch = 0.25×0.125 mm
- **0201**: 0.02×0.01 inch = 0.6×0.3 mm
- **0402**: 0.04×0.02 inch = 1.0×0.5 mm
- **0603**: 0.06×0.03 inch = 1.6×0.8 mm
- **0805**: 0.08×0.05 inch = 2.0×1.25 mm

Metric Code (L × W in 0.1 mm):

- **0402**: 0.4×0.2 mm = 01005 Imperial
- **0603**: 0.6×0.3 mm = 0201 Imperial
- **1005**: 1.0×0.5 mm = 0402 Imperial
- **1608**: 1.6×0.8 mm = 0603 Imperial
- **2012**: 2.0×1.25 mm = 0805 Imperial

2.2 Complete Size Comparison Table

Imperial	Metric	Length (mm)	Width (mm)	Height (mm)	Power Rating	Typical Use
008004	0201	0.25	0.125	0.15	1/64 W	Next-gen wearables
01005	0402	0.4	0.2	0.2	1/32 W	Medical implants
0201	0603	0.6	0.3	0.25	1/20 W	Smartphones
0402	1005	1.0	0.5	0.35	1/16 W	Consumer electronics
0603	1608	1.6	0.8	0.45	1/10 W	General SMT
0805	2012	2.0	1.25	0.45	1/8 W	Power applications
1206	3216	3.2	1.6	0.55	1/4 W	Legacy / power

Table 1: SMD Passive Component Size Comparison (Imperial & Metric)

2.3 Visual Size Comparison

The image below shows the relative sizes of common SMD components from 1206 down to 01005, alongside metric dimensions for reference.

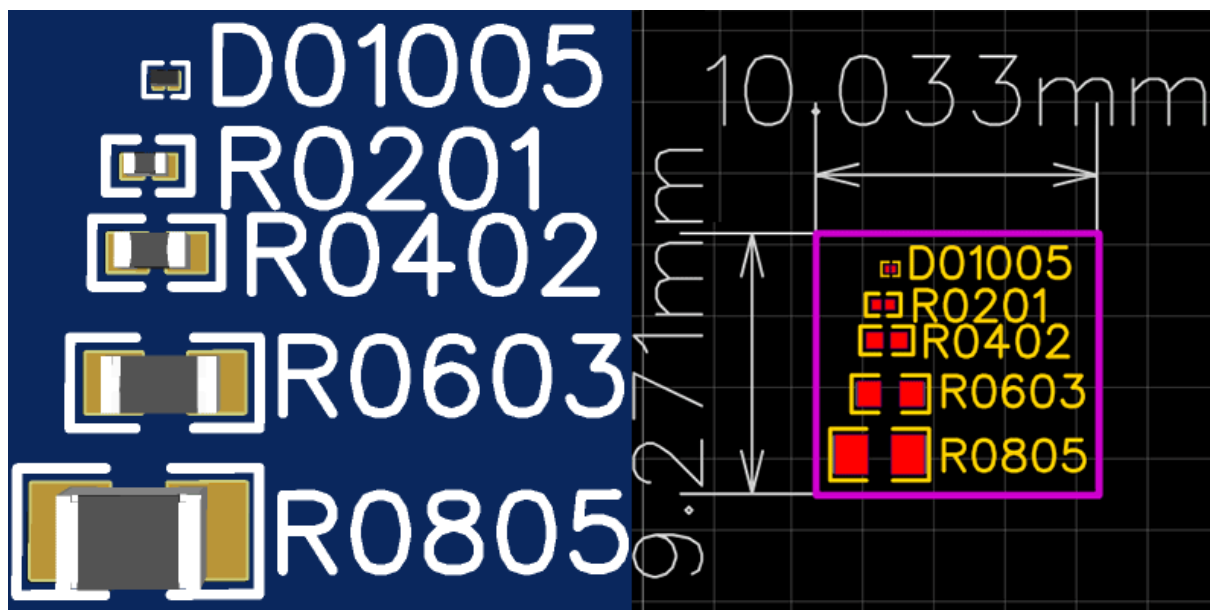


Figure 1: SMD Size Comparison — 01005 to 0805 with Dimensions

3. 01005 COMPONENTS (0.4 × 0.2 mm)

3.1 Overview

01005 (Metric 0402) is currently the **smallest widely used passive component** in high-volume PCBA manufacturing. At 0.4 × 0.2 mm, it is barely visible to the naked eye and requires specialized equipment and processes.

Physical Characteristics:

- **Length:** 0.4 mm (400 μm)
- **Width:** 0.2 mm (200 μm)
- **Height:** 0.2 mm (200 μm)
- **Weight:** ~0.04 mg
- **Terminal pad size:** ~0.1 × 0.2 mm
- **Comparison:** Smaller than a grain of sand; comparable to a human hair width

3.2 Assembly Challenges

01005 components present extreme challenges at every stage of assembly:

1. Solder Paste Printing

- **Stencil thickness:** 80–100 μm (thinner than standard 150 μm)
- **Aperture size:** 200 × 250 μm (laser-cut, electroformed, or stepped)
- **Paste type:** Type V (15–25 μm) or Type VI (5–15 μm) powder required
- **Printer accuracy:** ± 15 μm placement accuracy mandatory
- **Squeegee:** Metal blade with 45° angle, 5–10 kg pressure

2. Pick & Place

- **Nozzle:** Custom 0.3 mm or 0.4 mm ceramic nozzle
- **Vision system:** High-resolution camera (5–10 μm pixel size)
- **Placement accuracy:** ± 25 μm at 3σ
- **Placement force:** 0.5–1.0 N (excessive force damages component)
- **Feeder:** 8 mm tape with 2 mm pitch; vibration-free operation

3. Reflow Soldering

- **Profile:** Very tight process window; ± 5°C tolerance
- **Tombstoning:** Extremely high risk due to small thermal mass differential
- **Pad design:** Home plate or rounded rectangle to reduce tombstoning
- **Nitrogen atmosphere:** Strongly recommended for improved wetting
- **Peak temperature:** 235–245°C (lower than standard SAC305 to reduce defects)

4. Inspection

- **AOI:** 5–10 μm resolution camera required
- **X-ray:** Not applicable for passives; used for micro-BGA underfill verification
- **Microscope:** 40–100 $\times$ magnification for manual verification

■■ **Critical:** 01005 components are highly sensitive to electrostatic discharge (ESD). Handle only in ESD-controlled environments ($\leq 100\text{V}$) with grounded tools and wrist straps. Component damage from ESD may not be visible but causes latent failures.

4. 008004 COMPONENTS (0.25 × 0.125 mm)

4.1 Overview

008004 (Metric 0201) represents the **next frontier in component miniaturization**. At 0.25 × 0.125 mm, it is smaller than most grains of sand and pushes the boundaries of conventional SMT assembly. As of 2026, 008004 is used in limited high-volume applications (wearables, hearing aids, medical implants) and requires cutting-edge equipment.

Physical Characteristics:

- **Length:** 0.25 mm (250 µm)
- **Width:** 0.125 mm (125 µm)
- **Height:** 0.15 mm (150 µm)
- **Weight:** ~0.01 mg
- **Terminal pad size:** ~60 × 100 µm
- **Comparison:** 1/3 the size of 01005; 1/10 the size of 0201

4.2 Assembly Requirements

008004 requires equipment and processes beyond standard SMT capabilities:

1. Jet Dispensing (No Stencil)

- **Method:** Piezoelectric jet dispenser shoots solder paste droplets
- **Droplet size:** 80–150 µm diameter
- **Accuracy:** ± 10 µm
- **Paste:** Type VI (5–15 µm) or Type VII (2–11 µm)
- **Advantage:** No stencil required; flexible pattern programming

2. Advanced Pick & Place

- **Nozzle:** 0.2 mm ceramic or diamond-coated nozzle
- **Vision:** Sub-micron resolution with AI-assisted recognition
- **Accuracy:** ± 10 µm at 3σ
- **Force:** 0.1–0.3 N (ultra-light touch)
- **Feeder:** Custom waffle pack or 4 mm tape

3. Specialized Reflow

- **Oven:** 10+ zone with ± 2°C uniformity
- **Atmosphere:** Pure nitrogen (< 100 ppm O₂) mandatory
- **Profile:** Rapid ramp with minimal soak; peak 230–240°C
- **Conveyor:** Mesh belt to prevent vibration-induced displacement

4. Inspection

- **AOI:** 3–5 μm resolution with telecentric lens
- **3D SPI:** Solder paste volume measurement at sub-micron level
- **SEM:** Scanning Electron Microscope for failure analysis

■ **Industry Status (2026):** 008004 is not yet standard in most contract manufacturers. It is used primarily by Tier-1 OEMs (Apple, Samsung, medical device makers) with dedicated miniaturization lines. In-house capability for 008004 requires significant capital investment in jet dispensing and sub-micron placement systems.

5. MICRO-BGA & CHIP SCALE PACKAGES (CSP)

5.1 Overview

While passives shrink in two dimensions, active components (ICs) shrink in three dimensions through advanced packaging. **Micro-BGA** and **CSP (Chip Scale Package)** are the smallest active component formats, with package sizes approaching the die dimensions.

CSP Definition (IPC/JEDEC J-STD-012):

A CSP is a single-die, direct surface-mountable package with an area no more than 1.2× the area of the die, and a ball pitch no larger than 1.0 mm.

5.2 Common Micro-BGA / CSP Formats

Package	Ball Pitch	Ball Diameter	Package Size	I/O Count	Typical Use
WLCSP	0.35–0.5 mm	0.2–0.3 mm	1.0–3.0 mm	4–100	Mobile PMIC, RF
CSP (0.5mm)	0.5 mm	0.3 mm	3.0–5.0 mm	16–64	Memory, sensors
CSP (0.65mm)	0.65 mm	0.4 mm	4.0–7.0 mm	24–100	Microcontrollers
CSP (0.8mm)	0.8 mm	0.5 mm	5.0–10.0 mm	48–144	FPGA, DSP
Micro-BGA	0.4–0.5 mm	0.25–0.3 mm	2.0–4.0 mm	16–49	Camera modules

Table 2: Micro-BGA & CSP Package Formats

5.3 WLCSP (Wafer-Level Chip Scale Package)

WLCSP is the ultimate miniaturization format — the package is created directly on the silicon wafer before dicing. There is no substrate or leadframe; solder balls are placed directly on the die passivation.

Key Characteristics:

- **Size:** Identical to die size (1.0–3.0 mm typical)
- **Height:** 0.3–0.5 mm (including solder balls)
- **Ball pitch:** 0.35–0.5 mm
- **Ball diameter:** 0.2–0.3 mm
- **I/O density:** Very high — up to 100+ I/O in 3 mm²

Assembly Challenges:

- **Fragile:** No substrate protection — die is exposed to mechanical stress
- **Underfill mandatory:** LOCTITE 3549 or equivalent required for reliability
- **PCB pad design:** Solder mask defined (SMD) pads preferred
- **Reflow:** Nitrogen mandatory; peak 240–250°C
- **Inspection:** X-ray for voiding; C-SAM for underfill verification

5.4 CSP vs. Conventional BGA Comparison

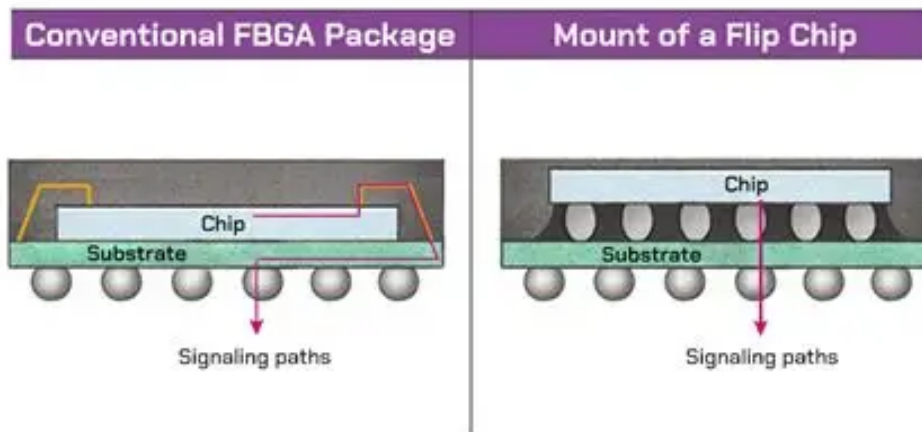


Figure 2: Conventional FBGA vs. Flip Chip CSP — Signaling Path Comparison

6. SPECIALIZED PROCESS REQUIREMENTS

6.1 Solder Paste for Miniature Components

Standard Type III paste (25–45 μm) is too coarse for 01005 and 008004. Smaller powder sizes are mandatory.

Component	Powder Type	Particle Size	Stencil / Method	Flux Type
0201	Type IV	20–38 μm	Laser stencil 80–100 μm	No-clean ORH0
01005	Type V	15–25 μm	Electroformed stencil 60–80 μm	No-clean ORH0
008004	Type VI/VII	5–15 μm	Jet dispensing (no stencil)	No-clean ORH0
Micro-BGA 0.4mm	Type V	15–25 μm	Laser stencil 100 μm	No-clean ORH1
WLCSP	Type V	15–25 μm	Laser stencil 100 μm	No-clean ORH1

Table 3: Solder Paste Requirements by Component Size

6.2 PCB Design Rules

Miniature components require specialized PCB design to ensure manufacturability.

Pad Design (01005 / 008004):

- **Pad length:** 0.2–0.25 mm (slightly longer than component for solder fillet)
- **Pad width:** 0.15–0.2 mm
- **Pad shape:** Home plate or rounded rectangle (reduces tombstoning)
- **Solder mask opening:** 0.05 mm larger than pad on all sides
- **Trace width:** ≤ 0.1 mm (4 mil) to minimize thermal differential

Micro-BGA / CSP Pad Design:

- **Pad diameter:** 0.2–0.25 mm (for 0.4–0.5 mm pitch)
- **Pad type:** Solder mask defined (SMD) preferred over non-solder mask defined (NSMD)
- **Via-in-pad:** Avoid for 0.4 mm pitch; use dog-bone routing if necessary
- **Trace escape:** 0.075 mm (3 mil) lines and spaces minimum

6.3 Reflow Profile Modifications

Miniature components require modified reflow profiles to prevent defects.

For 01005 / 008004 Passives:

- **Preheat ramp:** 1.0–1.5°C/sec (slower than standard to reduce thermal shock)
- **Soak zone:** Minimize or eliminate (reduces tombstoning)
- **Peak temperature:** 235–245°C (lower than standard 250°C)
- **Time above liquidus:** 30–45 seconds (shorter than standard)

- **Cooling rate:** 2–4°C/sec (controlled cooling reduces grain growth)

For Micro-BGA / CSP:

- **Preheat ramp:** 1.5–2.0°C/sec
- **Soak:** 150–180°C for 60–90 seconds (activates flux, reduces voids)
- **Peak:** 245–250°C (standard SAC305)
- **TAL:** 60–90 seconds (longer for large thermal mass)
- **Nitrogen:** < 100 ppm O₂ mandatory for 0.4 mm pitch

7. COMMON DEFECTS & SOLUTIONS

7.1 Tombstoning (Drawbridging)

The most common defect for 0201, 01005, and 008004. One end of the component lifts off the pad and stands vertically.

Causes:

- Unequal wetting forces on the two terminations
- Uneven solder paste deposition (volume or placement)
- Excessive soak time causing oxidation of one pad
- Component body shadowing one pad from reflow heat

Solutions:

- **Pad design:** Home plate shape with reduced pad length
- **Paste:** Reduce aperture by 10% to reduce solder volume
- **Profile:** Eliminate soak zone; use rapid ramp to peak
- **Nitrogen:** Reduces oxidation differential between pads
- **Placement:** Ensure centered placement within $\pm 25 \mu\text{m}$

7.2 Bridging

Solder bridges between adjacent pads or components, causing short circuits.

Causes:

- Excessive solder paste volume
- Misaligned stencil print
- Component placement offset
- Insufficient solder mask between pads

Solutions:

- **Stencil:** Reduce aperture 1:1 to pad size (no oversize)
- **Print:** Verify $\pm 15 \mu\text{m}$ print accuracy
- **Placement:** Tighten placement tolerance to $\pm 15 \mu\text{m}$
- **Solder mask:** Minimum 0.075 mm (3 mil) web between pads

7.3 Missing / Skipped Components

Component not present on the board after reflow.

Causes:

- Component not picked (feeder empty or misfeed)
- Component dropped during transport (static or vibration)

- Component stuck in nozzle (double pick)
- Component blown off by reflow oven convection

Solutions:

- **Feeder:** Verify tape advance and cover tape peel
- **Nozzle:** Clean and inspect for wear; use vacuum verification
- **Placement force:** Calibrate to 0.5–1.0 N for 01005
- **Oven:** Reduce convection velocity; use mesh belt

7.4 Head-in-Pillow (HiP) — Micro-BGA

BGA solder ball partially collapses but does not fully bond to PCB pad.

Causes:

- PCB or BGA warpage during reflow
- Oxidation on ball or pad surface
- Insufficient flux activity
- Non-wetting due to contamination

Solutions:

- **Profile:** Reduce peak temperature; extend TAL
- **Support:** Use board support pins to reduce warpage
- **Flux:** Use higher activity no-clean flux (ORH1)
- **Storage:** Bake BGA at 125°C for 24 hours before use
- **Nitrogen:** Mandatory for 0.4 mm pitch

■ **In-House Tip:** For 01005 production, implement 100% AOI with 5 µm resolution. Defect rates for 01005 are typically 500–2000 PPM without optimized processes. With proper pad design, Type V paste, nitrogen reflow, and tight placement, defect rates can be reduced to < 100 PPM.

8. EQUIPMENT REQUIREMENTS SUMMARY

Handling the smallest components requires investment in specialized equipment across all assembly stages.

Process Stage	0201	01005	008004	Micro-BGA
Solder Paste	Type IV Laser stencil	Type V Electroformed stencil	Type VI/VII Jet dispensing	Type V Laser stencil
Printer Accuracy	± 20 µm	± 15 µm	± 10 µm	± 20 µm
Placement Accuracy	± 35 µm	± 25 µm	± 10 µm	± 25 µm
Vision Resolution	5 µm	3 µm	1 µm	3 µm
Reflow Atmosphere	Air acceptable	N ² recommended	N ² mandatory	N ² mandatory
AOI Resolution	7 µm	5 µm	3 µm	5 µm
X-Ray	Not required	Not required	Not required	Required
3D SPI	Recommended	Required	Required	Required

Table 4: Equipment Requirements by Component Size

9. QUICK REFERENCE SUMMARY

Smallest Passives:

- **008004:** 0.25×0.125 mm — Next-gen, jet dispensing required
- **01005:** 0.4×0.2 mm — Current smallest high-volume standard
- **0201:** 0.6×0.3 mm — Standard for smartphones
- **0402:** 1.0×0.5 mm — General compact electronics

Smallest Actives:

- **WLCSP:** 1.0–3.0 mm die-size package, 0.35–0.5 mm ball pitch
- **Micro-BGA:** 2.0–4.0 mm, 0.4–0.5 mm pitch, 16–49 I/O
- **CSP:** $\leq 1.2 \times$ die size, 0.5–0.8 mm pitch

Paste Requirements:

- **0201:** Type IV (20–38 μm), laser stencil 80–100 μm
- **01005:** Type V (15–25 μm), electroformed stencil 60–80 μm
- **008004:** Type VI/VII (5–15 μm), jet dispensing
- **Micro-BGA:** Type V (15–25 μm), laser stencil 100 μm

Placement Requirements:

- **0201:** ± 35 μm , 5 μm vision, ceramic nozzle
- **01005:** ± 25 μm , 3 μm vision, 0.3 mm nozzle
- **008004:** ± 10 μm , 1 μm vision, 0.2 mm nozzle
- **Micro-BGA:** ± 25 μm , 3 μm vision, underfill mandatory

Reflow Requirements:

- **Passives (01005/008004):** Slow ramp, no soak, peak 235–245°C, N $\blacksquare$ recommended/mandatory
- **Micro-BGA/CSP:** Standard profile, peak 245–250°C, N $\blacksquare$ mandatory
- **WLCSP:** Underfill (LOCTITE 3549) mandatory after reflow

Key Defects & Prevention:

- **Tombstoning:** Home plate pads, reduced paste, no soak, N $\blacksquare$
- **Bridging:** 1:1 stencil aperture, tight placement, adequate solder mask
- **Missing:** Feeder verification, vacuum check, reduced oven convection
- **Head-in-Pillow:** Board support, bake BGA, extend TAL, N $\blacksquare$

ESD & Handling:

- **All miniature components:** ESD-controlled environment $\leq 100\text{V}$
- **Handling tools:** Grounded tweezers, wrist straps, conductive mats
- **Storage:** Moisture-barrier bags with desiccant and HIC

References & Standards:

- IPC-7093: Design and Assembly Process Implementation for Bottom Termination Components
- IPC-7527: Stencil Design Guidelines
- IPC/JEDEC J-STD-012: Implementation of Flip Chip and Chip Scale Technology
- IPC-A-610: Acceptability of Electronic Assemblies
- IPC/J-STD-001: Requirements for Soldered Electrical and Electronic Assemblies
- JEDEC MO-252: 01005 Component Outline
- JEDEC MO-255: 008004 Component Outline
- IPC-TM-650: Test Methods Manual

End of Document — Smallest Components in PCBA Assembly Complete Guide